Biochemistry Cumulative Examination

Data Analysis

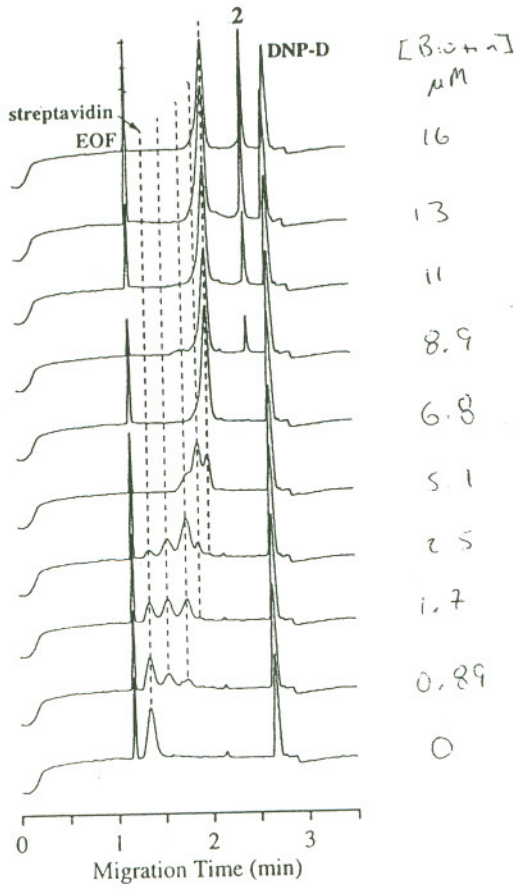
February 11, 1995

Recently C. Walsh and coworkers published a paper (*Biochem.* **1994**, *33*, 10616-10621) "Using affinity capillary electrophoresis to determine binding stoichiometries of protein-ligand interactions." There are surprisingly few ways to determine binding stoichiometry so a new method is welcome. Capillary electrophoresis (CE) is a fairly new technique for separating small amounts of material very efficiently (around 10^6 theoretical plates!). You do not need to worry about the details of how CE works just be aware that it separates compounds (and complexes) very efficiently. The interaction we are dealing with here is the interaction of a protein, R, with some number, n, of ligands, L with some characteristic disassociation constant, K_d. $R{\cdot}L_n \leftrightarrow R + nL$

Consider and analyze the following data to explain the questions below.

The first figure shows CE runs of streptavidin and biotin, which is bound extremely tightly ($K_d = 4 \times 10^{-14}$ M) by this protein. In this series of experiments the concentration of streptavidin in the sample was held constant at 1.7 μM while the concentration of biotin (labeled as compound **2**) was varied. The peak labeled DNP-D is an internal integration standard and the peak labeled EOF is the injection solvent front so both may be ignored.

[Biotin] μM	Normalized Area of Biotin Peak
0	0.00
0.89	0.00
1.7	0.00
2.5	0.00
5.1	0.00
6.8	0.00
8.9	0.21
11	0.38
13	0.62
16	0.85



1. What is the binding stoichiometry of biotin to streptavidin? Briefly explain your reasoning.

2. Is there any evidence above that would let you say if the binding of biotin is an all or none process or can the binding sites be only partially filled? If not, suggest an experiment to determine this point.

3. What complications might arise if you were studying a protein-ligand system that bound only weakly? Suggest a method or methods that would allow you to circumvent these problems.